1 Main Command Center Dashboard & Global Navigation Overview

The **ECOPROSUS Command Center** acts as the central monitoring bridge across distributed wireless edge nodes (ESP32-01 and ESP32-02). It provides a macro-level geographic overview across deployed sensing stations while continuously consolidating real-time telemetry into responsive visual gauges. At the bottom of the interface, instant radial readouts reflect current **Soil Moisture (32%)**, **Ambient Temperature (30°C)**, and **Relative Humidity (55%)**.

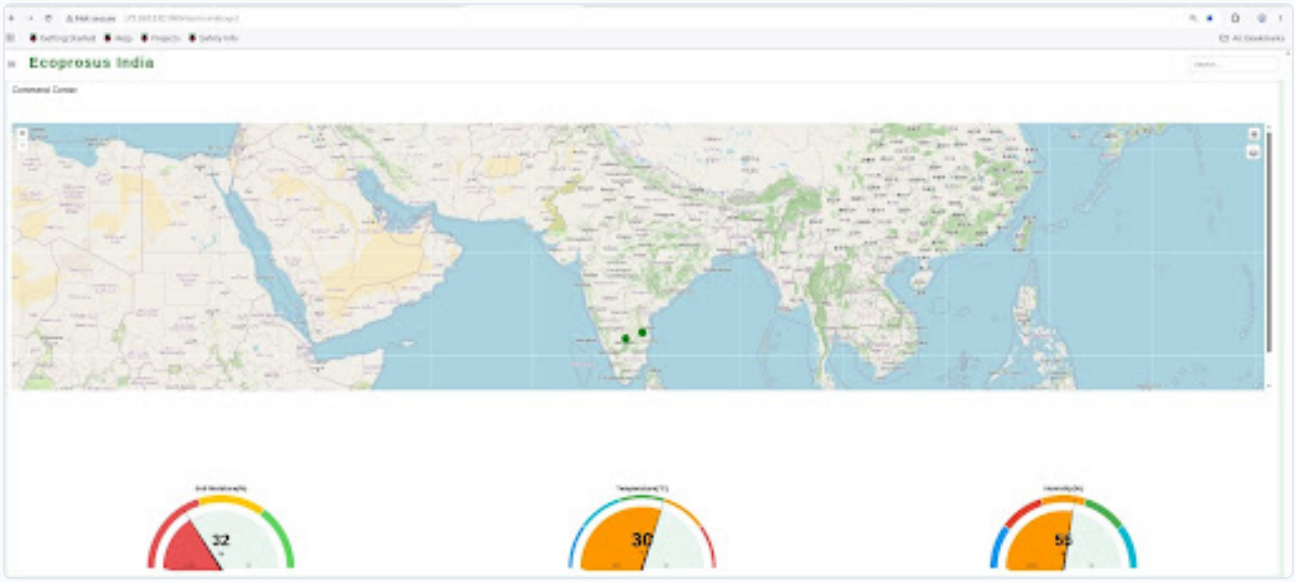


Figure 1: Main Command Center dashboard displaying the India geospatial distribution and synchronized telemetry gauges.

> **Global Navigation Architecture ("Ecoprosus"):** Regardless of which deep-dive telemetry view an operator is inspecting (Soil Moisture, Temperature, or Humidity), clicking the persistent **"Ecoprosus"** logo in the top-left navigation banner immediately resets the view back to this primary **Command Center** page.

Telemetry Parameter	Source Hardware Node	Baseline Gauge Range	Primary Command Center Function
Geospatial Map	OpenStreetMap / Leaflet Service	National Coverage (India)	Visual tracking of edge deployment sites & active connection states.
Soil Moisture (%)	ESP32 Node 01 (FC-28 Probe)	0% – 100% (Inverse Volumetric)	Substrate saturation level for automated irrigation triggers.
Temperature (°C)	ESP32 Node 02 (DHT11 Sensor)	0°C – 50°C (Standard Scale)	Ambient canopy thermal monitoring for micro-climate profiling.
Relative Humidity (%)	ESP32 Node 02 (DHT11 Sensor)	0% – 100% RH Range	Vapor pressure deficit & transpiration assessment.

2 Geospatial Node Clustering, Live Popups & Deep Routing

Zooming into specific deployment zones—such as the Bengaluru agricultural technology sector—reveals active GPS-tagged edge nodes indicated by dynamic green status markers. When an operator clicks any marker on the map, an interactive data popup opens immediately, displaying vital node diagnostic metadata including **Device Designation (ESP02 - Humidity)**, **Connection Status (ONLINE)**, and the **Current Real-Time Metric (69%)**.

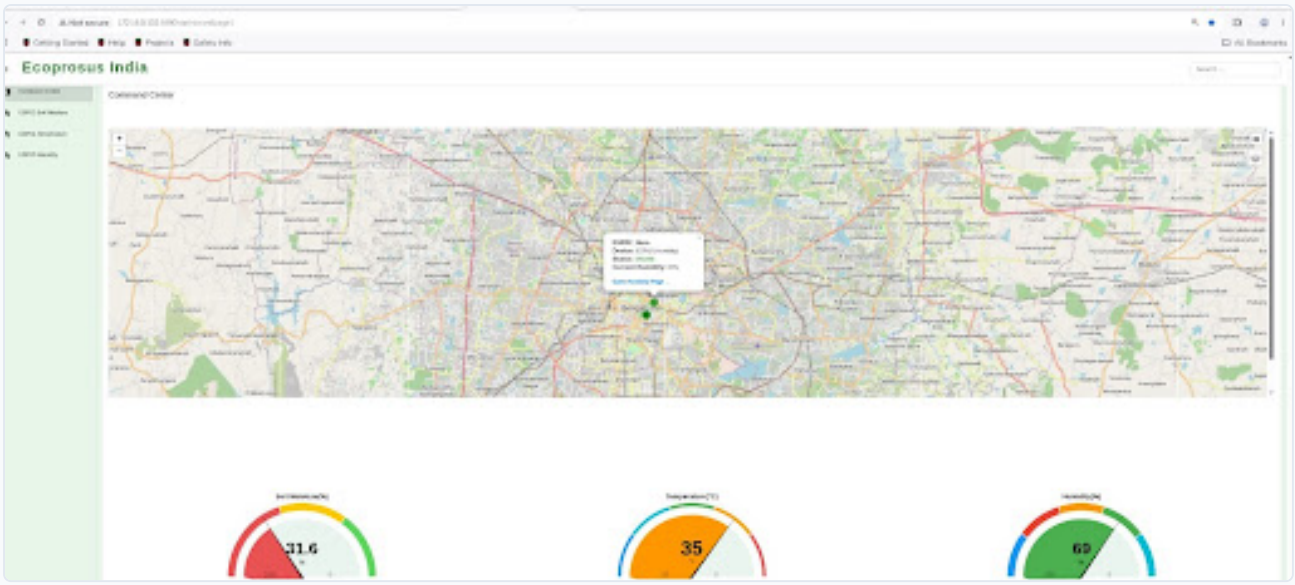


Figure 2: Interactive geospatial marker popup showing live station status and the dedicated routing hyperlink.

> **Direct Page Entry via Map Hyperlink:** Within the map popup, clicking the blue hyperlink (e.g., *"Open Humidity Page →"*) automatically routes the user directly into the dedicated telemetry view for that sensor, providing instant access to full historical trends and raw logs without manual sidebar navigation.

Interactive Map Marker Diagnostics

Green circular markers indicate continuous heartbeat reception. If MQTT telemetry times out beyond 60 seconds, marker styling shifts state to alert maintenance engineers of field offline events.

Sub-Menu Navigation Architecture

The collapsible left-hand navigation drawer allows manual selection between `Command Center`, `ESP01-Soil Moisture`, `ESP02-Temperature`, and `ESP02-Humidity`.

Synchronized Gauge Reflection

When drilling down into local cluster nodes, the bottom gauge cluster dynamically updates to highlight current localized readings: Soil Moisture (31.6%), Temperature (35°C), and Humidity (69%).

Fast Return Flow

Operators can transition smoothly from deep node troubleshooting back to the global overview simply by clicking **Ecoprosus** on the top masthead.

3 Soil Moisture Telemetry Analytics (ESP32 Node 01)

The dedicated **Soil Moisture** dashboard presents complete volumetric moisture telemetry derived from the FC-28 analog resistance probe on GPIO 34. The page is structured into three synchronized diagnostic tiers: a multi-band dial gauge indicating real-time moisture, a continuous time-series trend line, and a historical data logging table displaying exact UTC/IST-calibrated transmissions.

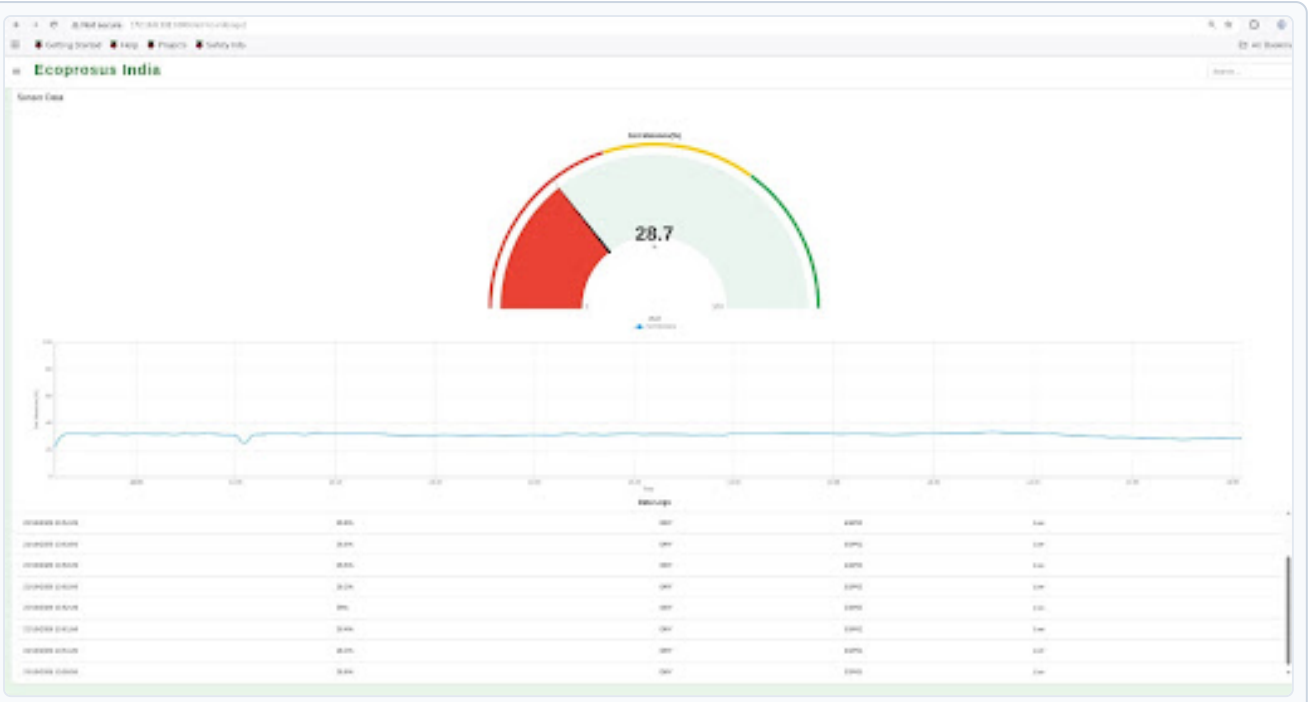


Figure 3: Dedicated Soil Moisture analytics page showing dial gauge (28.7%), historical trend chart, and event log table.

Sensor Condition Triaging: The gauge utilizes color-coded thresholds where red designates arid conditions (<30%), yellow designates transition/sub-optimal hydration (30%–60%), and green indicates field capacity. The table explicitly tags status as DRY when ADC voltage rises near saturation.

Timestamp (IST Recorded)	Moisture Value (%)	Condition State	Transmitting Device	Telemetry Status
22/10/2026 13:54:26	28.8%	DRY	ESP01	LIVE (Double-Blink TX Confirmed)
22/10/2026 13:53:56	28.5%	DRY	ESP01	LIVE (ACK Verified)
22/10/2026 13:53:26	28.5%	DRY	ESP01	LIVE (ACK Verified)
22/10/2026 13:52:56	28.2%	DRY	ESP01	LIVE (ACK Verified)

4 Canopy Temperature Telemetry Analytics (ESP32 Node 02)

The **Temperature Telemetry** page provides precise environmental thermal analysis recorded by the DHT11 sensor integrated into ESP32 Node 02 on GPIO 19. A prominent radial gauge highlights current ambient canopy warmth at **26°C**, positioned comfortably within the green optimal vegetation band (20°C to 35°C). Blue lower zones indicate chilling risks while upper red arcs flag heat stress conditions exceeding 38°C.

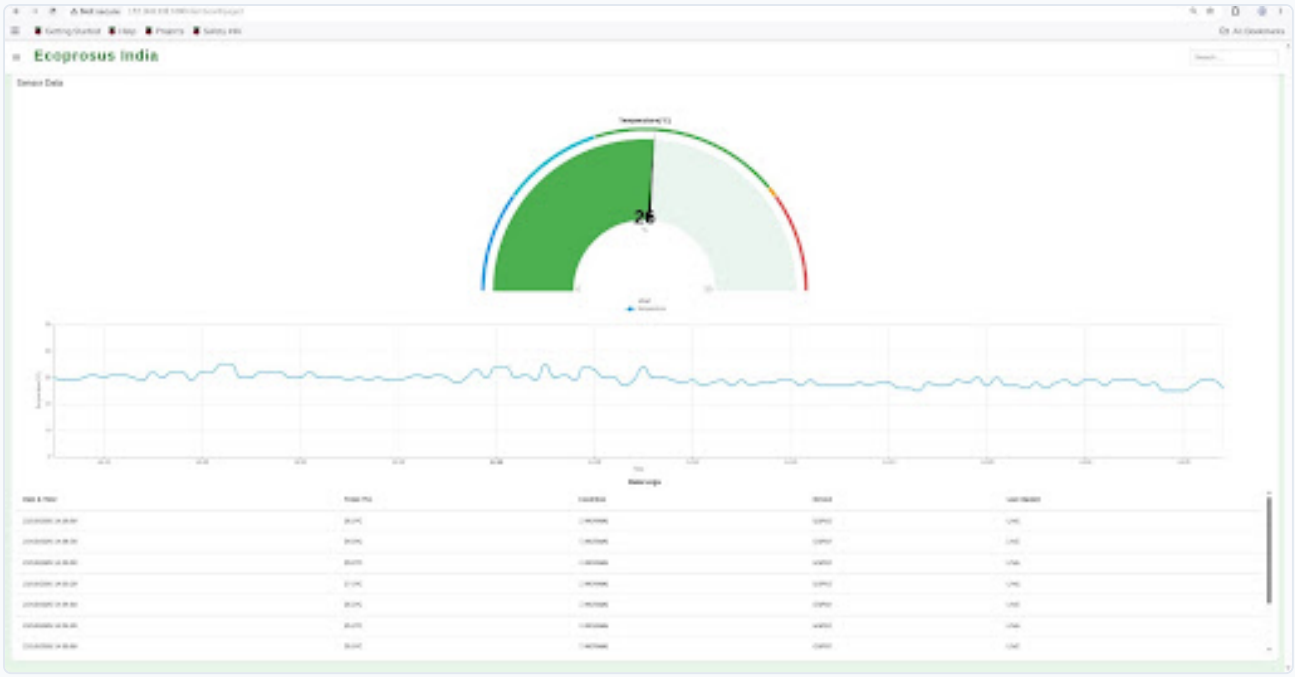


Figure 4: Temperature telemetry page showing center gauge (26°C), time-series chart, and historical record table.

Thermal Drift Monitoring: The continuous trend chart traces micro-climate temperature variations between 13:40 and 14:35, demonstrating rapid heat stabilization around 26.0°C–29.0°C following transient ventilation cycles.

Timestamp (IST Recorded)	Temperature (°C)	Operating Condition	Hardware Node	Telemetry Pipe
22/10/2026 14:36:58	26.0°C	NORMAL	ESP02	MQTT QoS 0 / Verified
22/10/2026 14:36:28	29.0°C	NORMAL	ESP02	MQTT QoS 0 / Verified
22/10/2026 14:35:58	29.0°C	NORMAL	ESP02	MQTT QoS 0 / Verified
22/10/2026 14:35:28	27.0°C	NORMAL	ESP02	MQTT QoS 0 / Verified

5 Relative Humidity Telemetry Analytics (ESP32 Node 02)

Relative atmospheric humidity is crucial for calculating vapor pressure deficits (VPD) and preventing crop fungal outbreaks. The **Humidity Telemetry** dashboard displays real-time capacitive readings streamed by Node 02. The large central dial gauge reflects a current moisture level of **51% RH**, supported by a granular time-series line chart tracking humidity dynamics and an authoritative historical log table.

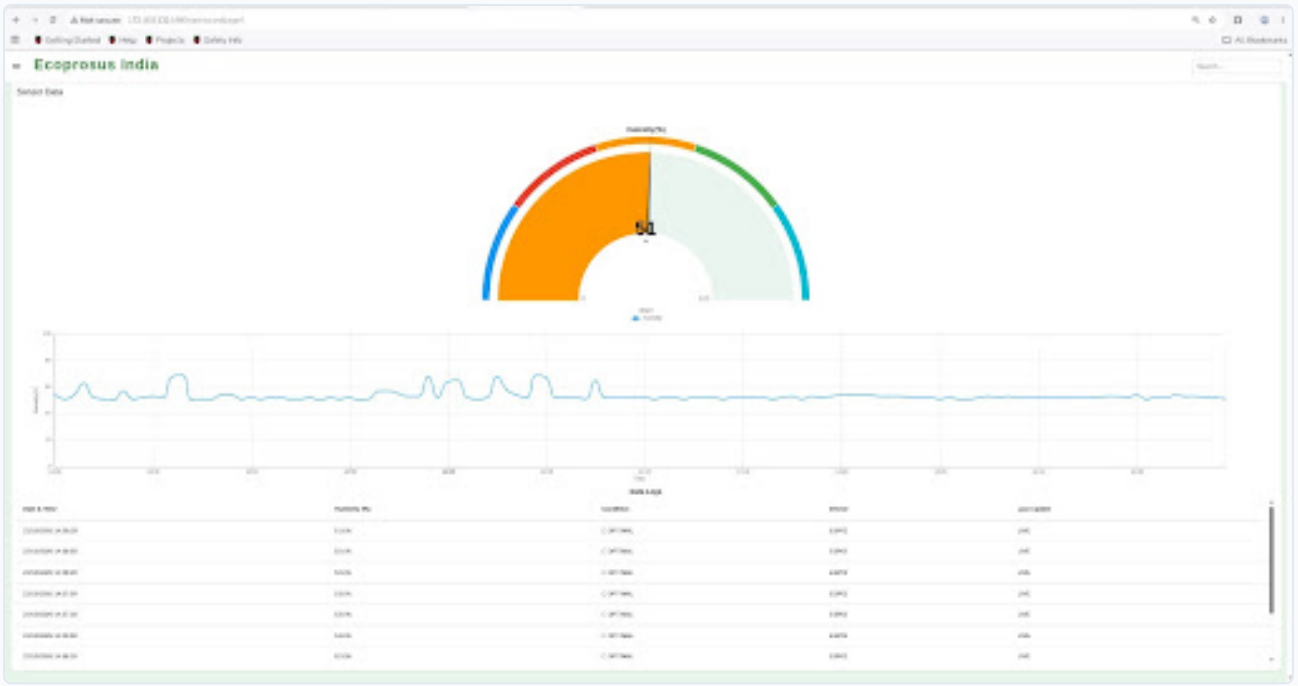


Figure 5: Dedicated Humidity dashboard featuring gauge (51% RH), stability curve, and log verification table.

> **Agronomic Optimum Tracking:** The 40% to 70% RH green zone represents the target comfort boundary for agricultural greenhouses. The log table verifies stable regulation around 51.0%–54.0%, classifying each packet as `OPTIMAL`.

Timestamp (IST Recorded)	Humidity Value (% RH)	Classified State	Transmitting Station	Telemetry Pipe
22/10/2026 14:39:28	51.0%	OPTIMAL	ESP02	LIVE (NTP Synced)
22/10/2026 14:38:58	52.0%	OPTIMAL	ESP02	LIVE (NTP Synced)
22/10/2026 14:38:28	52.0%	OPTIMAL	ESP02	LIVE (NTP Synced)
22/10/2026 14:37:58	53.0%	OPTIMAL	ESP02	LIVE (NTP Synced)

6 Node-RED Backend Flow & Bidirectional Pipeline Architecture

The core intelligence and data routing of the ECOPROSUS architecture resides inside the **Node-RED Visual Flow Engine** hosted on the Raspberry Pi gateway (172.16.0.132:1880). The backend ingests asynchronous MQTT JSON packets, bifurcates payloads into scalar values, updates UI Dashboard 2.0 widgets, renders the geographic Leaflet map, and generates immediate bidirectional MQTT acknowledgment handshakes.



Figure 6: Complete Node-RED Flow 1 diagram detailing MQTT input nodes, parser functions, UI widgets, and ACK return pipes.

Closed-Loop ACK Generation Logic: Upon receiving valid telemetry from `ecoprosus/esp32-01/soil/moisture` or Node 02's topic, dedicated function nodes (`Soil ACK` / `Temp ACK`) fire a string `"ACK"` back to the respective ACK topics, triggering hardware verification LEDs on the physical boards.

MQTT Ingest & JSON Decomposition

Input nodes (`ESP32-01 Soil` & `ESP32-02 Temp Humidity`) deserialize JSON strings into JavaScript objects, feeding gauge and chart nodes concurrently.

Geographic Map Ingestion Pipeline

Function nodes (`Map - Soil`, `Map - Temp`, `Map - Hum`) package GPS coordinates and current metrics into the `India Map Overview` worldmap node.

Tabular History Buffering

Table-formatting sub-flows manage bounded FIFO arrays for logging widgets, ensuring that historical records persist without consuming excessive browser memory.

Dashboard 2.0 Sidebar Grouping

The right sidebar organizes UI groups (Command Center, ESP01-Soil Moisture, ESP02-Temperature, and ESP02-Humidity) matching the production layout hierarchy.